ANRAJ SONI

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EDUCATION

University of British Columbia (UBC), Vancouver, BC Canada — Expected Graduation: April 2027
Mechanical Engineering — *Dean's Honour List*: 2022, 2023 & 2026

ETH Zürich, Zürich, Switzerland — Exchange Semester
Robotics/Embedded Systems — February - June 2026

EXPERIENCE

Arc Boats — Los Angeles, CA
Test Engineering Intern — May - August 2025

- Joined as the first Test Engineering Intern, contributing to the validation of Arc's first-generation electric boats
- Designed and executed structural validation of stern thruster and bow tube, reducing 25 labor hours per boat
- Developed automated QR code inventory system to track critical equipment, saving US$4,000+ annually
- Implemented RTK GPS test asset, providing cm-level accuracy for heading, position, and speed through CAN signals

Rocket Lab — Toronto, ON
Mechanical Engineering Intern — January - April 2025

- Created rotor/stator gauges for reaction wheels, reducing verification time from 10 minutes to 15 seconds per part
- Conducted design trade studies for star tracker baffles to optimize manufacturability and performance
- Redesigned parts of reaction wheel to improve manufacturability, notably changing swage pins to solder sockets

Tesla, Inc. — Sunnyvale, CA
Hardware Test Engineering Intern — September - December 2024

- Built thermal test rig to validate pump performance on the Semi truck, integrating with vehicle CAN data signals
- Diagnosed root cause of NVH issues for Model 3/Y cabin temperature sensors; recommended countermeasures projected to save over US$300,000+ for future vehicle programs
- Conducted HVAC compressor efficiency testing of oil/refrigerant interaction in extreme conditions, providing data on operation limits across Model 3/Y/S/X

AtkinsRéalis Group Inc. — Mississauga, ON
Project Engineering Intern — May - August 2024

- Supported North America's first fleet of small modular nuclear reactors by tracking engineering deliverables, scheduling, leading interdisciplinary meetings with 10+ technical managers, and calculating SPI/CPIs, allowing project to progress from 30% to 60% completion

Parkland Corporation — Burnaby, BC
Designs Engineering Intern — May - December 2023

- Performed engineering calculations on 25+ pressure vessels in accordance with ASME BPVC to determine both minimum pressurization temperature or adequate torque values for bolted flange joints
- Led root cause analysis on failed heat exchangers, developing 5 Whys to identify contributing factors & implementing preventative maintenance actions to save over CA$100,000+ annually

DESIGN TEAM

UBC AeroDesign — Vancouver, BC
Fuselage Structures Lead — January 2022 – May 2024

- Led a team of 7 undergraduate students in design process of an aircraft fuselage from conceptualization to manufacturing, leading to 7th place finish out of 18 teams in the international 2024 SAE AeroDesign competition
- Developed model of aircraft with 40+ components in SolidWorks, ensuring adequate DFM/DFA principles
- Conducted analyses on all phases of flight with consideration for different configurations, static stability, structural sizing, and cost through fundamental solid mechanics principles like moment balancing, beam bending, and torsion
- Manufactured various components of aircraft using laser cutting, water jetting, composite wet layup, CNC machining, FDM 3D printing, hot wire cutting, spot welding, and basic hand tools

SKILLS

- **Mechanical:** FDM 3D Printing, Laser Cutting, Water Jetting, Carbon Fibre Manufacturing, Sheet Metal Design, CNC Machining, Milling, Lathing, Hand Tools
- **Electrical/Software:** SolidWorks, AutoCAD, MATLAB, Soldering, Arduino UNO, C/C++, PCAN Explorer, DAQ Systems, Oscilloscopes, Motor Control, Finite Element Analysis
- **Technical Knowledge:** Design for X, DFMEA/PFMEA, Material Selection, GD&T, DFM/DFA, Bill of Materials, Engineering Drawings, Continuous Improvement, Root Cause Analysis